

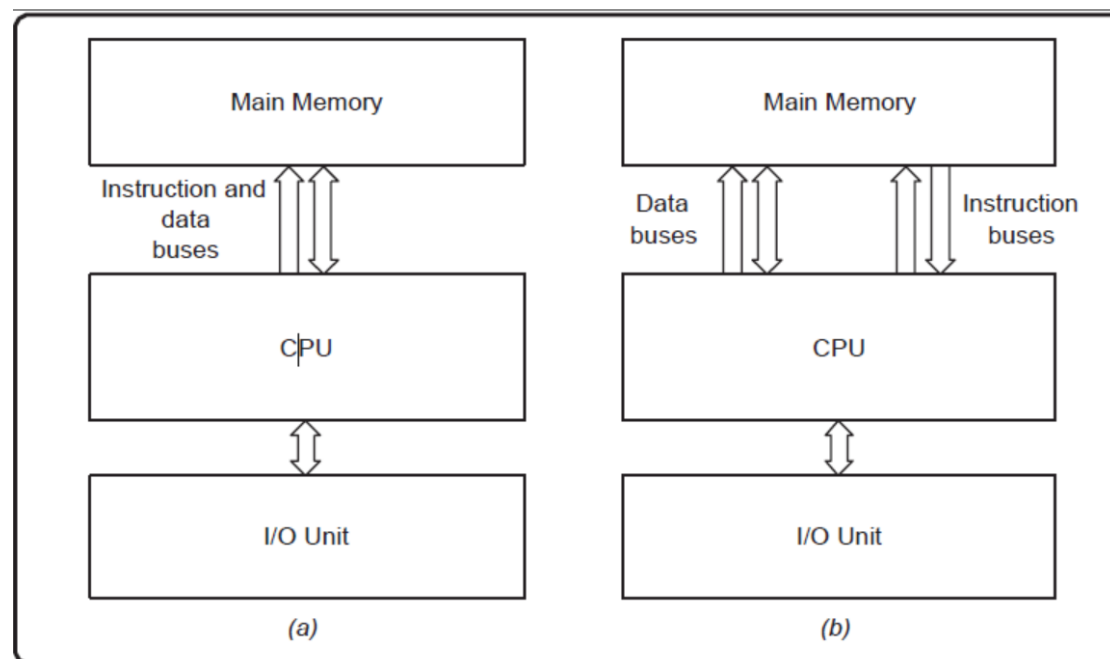
The Design of Computer Systems and Components

What is Computer Architecture?

Computer Architecture is the design or blueprint of a computer system that shows how all its parts work together to perform tasks, especially processing data. It focuses on the structure and interaction of components rather than the low-level technical details of how they are built.

The goal of computer architecture is to achieve a balance between performance, cost, power efficiency, and reliability. It focuses on designing components that work together smoothly such as the processor, memory units, buses, and control circuits to execute instructions, manage data, and produce accurate results.

In simple terms, computer architecture explains how a computer is built and how it works internally, from fetching and decoding instructions to executing them and displaying output to the user.

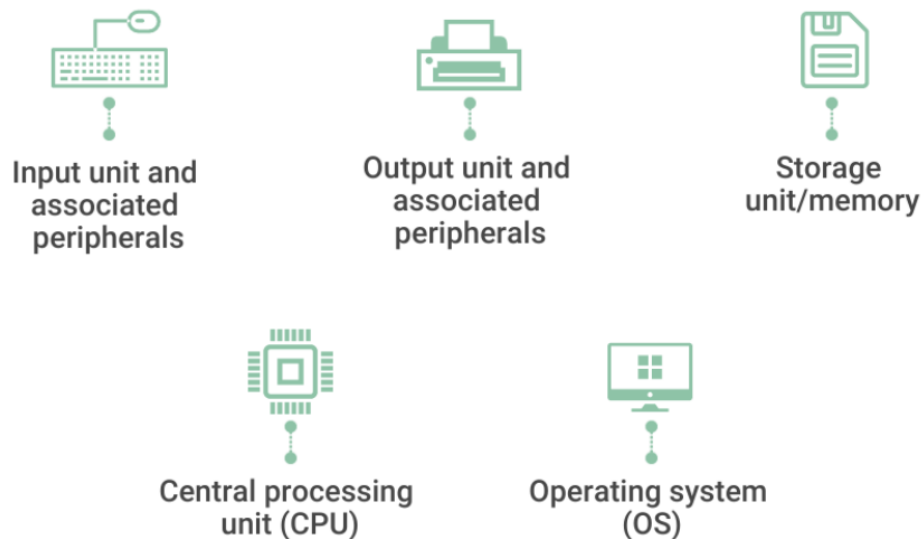


Components of Computer Architecture

Depending on the method of categorization, the parts of a computer architecture can be subdivided in several ways. The main components of a computer architecture are the CPU, memory, and peripherals. All these elements are linked by the system bus, which comprises an address bus, a

data bus, and a control bus. Within this framework, the computer architecture has eight key components, as described below.

Components of Computer Architecture



1. Input unit and associated peripherals

The input unit provides external data sources to the computer system. Therefore, it connects the external environment to the computer. It receives information from input devices, translates it to machine language, and then inserts it within the computer system. The keyboard, mouse, or other input devices are the most often utilized and have corresponding hardware drivers that allow them to work in sync with the rest of the computer architecture.

2. Output unit and associated peripherals

The output unit delivers the computer process's results to the user. A majority of the output data comprises music, graphics, or video. A computer architecture's output devices encompass the display, printing unit, speakers, headphones, etc.

To play an MP3 file, for instance, the system reads a number array from the disc and into memory. The computer architecture manipulates these numbers to convert compressed audio data to uncompressed audio data and then outputs the resulting set of numbers (uncompressed audio file)

to the audio chips. The chip then makes it user-ready through the output unit and associated peripherals.

3. Storage unit/memory

The storage unit contains numerous computer parts that are employed to store data. It is typically separated into primary storage and secondary storage.

Primary storage unit

This component of the computer architecture is also referred to as the main memory, as the CPU has direct access to it. Primary memory is utilized for storing information and instructions during program execution. Random access memory (RAM) and read-only memory (ROM) are the two kinds of memory:

- **RAM** supplies the necessary information straight to the CPU. It is a temporary memory that stores data and instructions intermittently.
- **ROM** is a memory type that contains pre-installed instructions, including firmware. This memory's content is persistent and cannot be modified. ROM is utilized to boot the machine upon initial startup. The computer is now unaware of anything outside the ROM. The chip instructs it on how to set up the computer architecture, conduct a power-on self-test (POST), and finally locate the hard drive so that the operating system can be launched.

Secondary storage unit

Secondary or external storage is inaccessible directly to the CPU. Before the CPU uses secondary storage data, it must be transferred to the main storage. Secondary storage permanently retains vast amounts of data. Examples include hard disk drives (HDDs), solid-state drives (SSDs), compact disks (CDs), etc.

4. Central processing unit (CPU)

The central processing unit includes registers, an arithmetic logic unit (ALU), and control circuits, which interpret and execute assembly language instructions. The CPU interacts with all the other parts of the

computer architecture to make sense of the data and deliver the necessary output.

1. Registers

These are high-speed and purpose-built temporary memory devices. Rather than being referred to by their address, they are accessed and modified directly by the CPU throughout execution. Essentially, they contain data that the CPU is presently processing. Registers contain information, commands, addresses, and intermediate processing results.

2. Arithmetic logic unit (ALU)

The arithmetic logic unit includes the electrical circuitry that performs any arithmetic and logical processes on the supplied data. It is used to execute all arithmetic (additions, subtractions, multiplication, division) and logical (<, >, AND, OR, etc.) computations. Registers are used by the ALU to retain the data being processed.

3. Control unit

The control unit collaborates with the computer's input and output devices. It instructs the computer to execute stored program instructions via communication with the ALU and registers. The control unit aims to arrange data and instruction processing.

The microprocessor is the primary component of computer hardware that runs the CPU. Large printed circuit boards (PCBs) are utilized in all electronic systems, including desktops, calculators, and internet of things (IoT) devices. The Intel 40004 was the first microprocessor with all CPU components on a single chip.

5. Bootloader

The firmware contains the bootloader, a specific program executed by the processor that retrieves the operating system from the disc (or non-volatile memory or network interface, as deemed applicable) and loads it into the memory so that the processor can execute it. The bootloader is found on desktop and workstation computers and embedded devices. It is essential for all computer architectures.

6. Operating system (OS)

The operating system governs the computer's functionality just above firmware. It manages memory usage and regulates devices such as the keyboard, mouse, display, and disc drives. The OS also provides the user with an interface, allowing them to launch apps and access data on the drive.

Typically, the operating system offers a set of tools for programs, allowing them to access the screen, disc drives, and other elements of the computer's architecture.

7. Buses

A bus is a tangible collection of signal lines with a linked purpose; a good example is the universal serial bus (USB). Buses enable the flow of electrical impulses between various components of a computer's design, transferring information from one system to another. The size of a bus is the count of information-transferring signal lines. A bus with a size of 8 bits, for instance, transports 8 data bits in a parallel formation.

In computer architecture, a bus is a communication pathway that transfers data, addresses, and control signals between different components of a computer system. The system bus mainly consists of three types of buses: the Address Bus, the Data Bus, and the Control Bus. These buses work together to ensure smooth communication between the CPU, memory, and input/output devices.

The **Address Bus** carries memory addresses from the CPU to memory or I/O devices, indicating where data should be read from or written to. It is usually unidirectional, meaning data flows only from the CPU to other components. The width of the address bus (for example, 16-bit, 32-bit, or 64-bit) determines how many unique memory locations a system can access. For instance, a 32-bit address bus can address up to 2^{32} distinct memory locations. Whenever the CPU needs to read or write data, it sends the specific address through the address bus so that the correct memory cell or device responds.

The **Data Bus** is responsible for carrying the actual data between the CPU, memory, and peripheral devices. Unlike the address bus, the data

bus is bidirectional, meaning data can travel both to and from the CPU. The width of the data bus defines how much data can be transmitted at one time. For example, a 64-bit data bus can transfer 64 bits of data simultaneously, which improves system performance and speed. When the CPU reads data from memory or writes data back to it, the information is transmitted through the data bus.

The **Control Bus** carries control and coordination signals that manage the actions of all connected components. It tells the system whether to perform a read or write operation, manages timing and synchronization, and handles interrupts and status signals. The control bus is generally bidirectional since it carries signals both from the CPU to devices and vice versa. Common control signals include “Read”, “Write”, “Clock”, and “Interrupt Request”. Without the control bus, the CPU would not be able to coordinate communication properly across the system.

Together, these three buses form the backbone of communication within a computer’s architecture.

8. Interrupts

Interrupts, also known as traps or exceptions in certain processors, are a method for redirecting the processor from the running of the current program so that it can handle an occurrence. Such an event might be a malfunction from a peripheral or just the fact that an I/O device has completed its previous duty and is presently ready for another one. Every time you press a key and click a mouse button, your system will generate an interrupt.